THE JOURNAL OF HISTOCHEMISTRY AND CYTOCHEMISTRY

Vol. 24, No. 1, pp. 100–111, 1976
Printed in U.S.A.

AN AUTOMATED MICROSCOPE FOR CYTOLOGIC RESEARCH A PRELIMINARY EVALUATION[1]

JOHN F. BRENNER, BROCK S. DEW, J. BRIAN HORTON, THOMAS KING, PETER W. NEURATH AND WILLIAM D. SELLES

Department of Therapeutic Radiology, Tufts-New England Medical Center, Boston, Massachusetts 02111 (J. F. B., J. B. H., P. W. N., W. D. S.); and Charles Stark Draper Laboratory, Cambridge, Massachusetts 02139 (B. S. D., T. K.)

A research-oriented system for automated microscopy is described from an operational point of view. The system consists of a microscope, a TV camera, an automatic cell finder and a servo-driven computer controlled stage. The system is interfaced to a NOVA 840 computer having 112,000 words of 16-bit core memory and extensive peripherals. It is capable of performing a wide variety of image processing tasks and is being used to study various aspects of automated microscopy, with applications in, but not limited to, cytology. Results of preliminary performance evaluations are given.

The primary objective of the Tufts-New England Medical Center image processing laboratory is the application of computer-assisted image analysis to problems of interest in the medical world. Especially in cytology, the goals of this research include the automation of existing procedures, such as differential white cell counting and chromosome karyotyping, as well as the making of new quantitative measurements that are not possible or practical by manual means. An example of the latter is the quantitative differentiation of lymphoblasts in acute lymphoblastic leukemia. Most of the research projects undertaken so far involve automated quantitative analysis of microscopic images. Accordingly, in collaboration with the Charles Stark Draper Laboratory, a research instrument was designed and built to study various aspects of automated microscopy. It is capable of performing a wide variety of the tasks that specialized automated microscopes of the future might have to perform. The instrument is called SCANIT (*S*ystem for *C*omputerized *AN*alysis of *I*mages at *T*ufts), and most of the relevant design considerations and construction details were reported by Dew *et al.* (2) in a previous issue of this journal. The present paper will briefly review the overall system from an operational rather than a design standpoint, pointing out several refinements and changes and, most importantly, will report on a preliminary evaluation of the instrument in actual use under research laboratory conditions.

[1]This work was supported in part by the U.S. Public Health Service under Grant GM21817 to the C.S. Draper Laboratory, and in part by the Health Services and Mental Health Administration under Grant HS00696 to Tufts-New England Medical Center.

OVERALL SYSTEM DESCRIPTION

Figure 1 is an overall view of the SCANIT system, which consists of a Leitz Dialux microscope, a servo-controlled automatic stage, a TV camera, a filter and illumination system, a video monitor, a push-button console and a dedicated 4800 Baud video terminal for operator communication. The computer system (not shown) to which SCANIT is interfaced is a Data General Nova 840 with 112,000 words of 16-bit core memory, 120 million bytes of interchangeable disc memory provided by two 60-million byte removable disc pack drives, 512,000 bytes of high speed fixed head disc storage, hardware floating point processor, industry compatible 9-track magnetic tape drive, 600 line/min printer, etc. The operating system used is "mapped" RDOS, revision 3, a *R*eal time *D*isc *O*perating *S*ystem supplied by Data General Corporation.

SCANIT is controlled by the computer in either a fully automatic or an interactive mode, the difference between the two modes being one of programming technique. For example, in the fully automatic mode, the system can be programmed to search a given area of a slide for leukocytes, center and focus each cell it finds, classify and count the cell and then resume the search. This procedure would continue without operator intervention until the designated search area had been completely covered, or

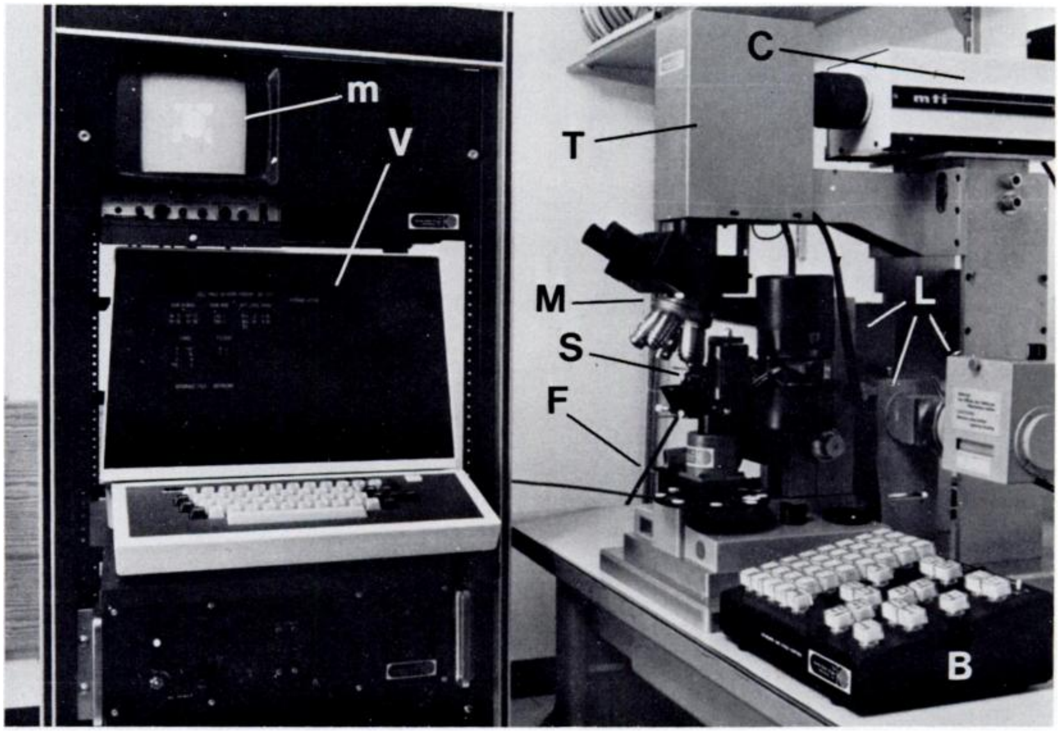

FIG. 1. Overall view of SCANIT system, showing the microscope (M), automatic stage (S), TV camera (C), rotary filter system (F), illumination system (L), video monitor (m), push-button console with manual stage controls (B), video terminal (V) and optical tower (T). NOVA 840 computer system not shown.

until a specified number of leukocytes had been counted. Finally, the number of cells in each category could be printed or displayed. This, of course, is the operation of an automatic differential white cell counter.

In an interactive mode, the system has been used to record and analyze leukocytes under operator control. Under this type of programming, the system waits until the operator has found and centered a particular cell. At the push of several buttons, the operator classifies, automatically focuses, and records six monochromatic images of the cell onto a disc pack, each image being recorded through a different narrow band interference filter. This mode is being used to record cell data for several current research projects.

The SCANIT system operates as a peripheral device, fully integrated into the RDOS operating system. A special, nonstandard device driver program and interrupt service routine were written to accomplish this. A "package" of Fortran-callable assembly language subroutines were also written to allow SCANIT to be controlled by standard Fortran IV programs. All developmental programming is done in Fortran IV, and assembly language is used only to speed up occasional compute-bound "number-crunching" routines. The RDOS operating system is used for all running, calibration and testing programs, except for a few stand-alone diagnostic programs used in testing the SCANIT interface. The real-time-interrupt handling and "task scheduling" capabilities of RDOS permit the debugging and running of other programs *simultaneously* with the operation of an interactive or automatic SCANIT program.

The automatic stage: The microscope stage, or "slide mover," is a unique feature of SCANIT. Designed and fabricated by the Draper Laboratory, it is a servo-actuated stage fully controled in X, Y and Z (focus) both manually and by computer. As can be seen from Figures 2 and 3, the slide is supported at the end of a double-pivoted arm, one pivot being in each of the drum-shaped enclosures. "X" is defined as the angle measured about the axis of the larger drum, while "Y" is the angle mea-

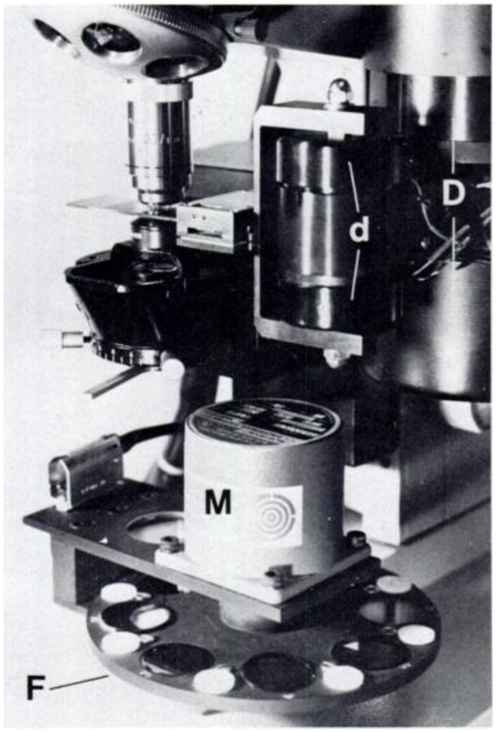

FIG. 2. Closeup of automatic stage and filter system. Large drum (D) contains both servo-resolver pairs, one above and one below the slide plane. The small drum (d) contains the electrically driven Z-axis motion effector. The entire stage is mounted in place of a standard Leitz stage, and therefore can also be moved in the Z-direction by the standard focus knobs. The filter disk (F) is driven directly by the computer-controlled stepper motor (M). Twenty-five steps are required to move between adjacent filters.

sured about the axis of the smaller drum. Both the X and Y servo-resolver pairs are in the larger drum, the X motion being driven directly and the Y motion being driven by a stainless steel band. The coordinates "X" and "Y" are thus biradial, not actually Cartesian, but the geometry of the arms gives a reasonable correspondence to Cartesian coordinates over the relatively small (16 x 16 mm) region of operation. The Z or focus motion is rectilinear and is effected by a solenoid and spring mechanism within the small drum.

The stage is analogue driven but digitally controlled in 0.25-μm steps in all axes. The X and Y range is ± 32768 steps, or ± 8.192 mm. A 16-bit signed binary integer thus specifies X or Y. The Z range is ± 128 steps, or ± 32 μm. A silicon photodiode, mounted in the optical tower above the trinocular tube of the microscope, is used in a cell finding system to detect relatively dark objects, *e.g.*, leukocyte nuclei, passing through a small (typically 8 μm in diameter) region of the visible field. The stage can be driven in a meander pattern by automatic hardware control, and the photodiode sensor is used to flag the position of leukocyte nuclei so that the stage can make controlled stops with these nuclei approximately centered in the field of view of the microscope. In this way, the stage handles the cell finding completely in hardware and signals the computer only when it has found and centered a cell.

The stage can also be used in a manual mode via push-button actuated electronic controls for X, Y and Z motion. In either manual or automatic mode, the digitized X, Y and Z positions of the slide can be read by the computer.

The scanner digitizer system: The heart of SCANIT is the scanner/digitizer system for converting the microscopic image into a set of numbers manageable by the computer. This is a specially designed television-based system, followed by an analogue-to-digital (A/D) converter to change the continuous video signal into discrete binary numbers.

The TV camera used is a modified MTI model 100, using an Amperex XQ1071B plumbicon. A plumbicon was selected over other tube types for its linearity (unity gamma), low lag, good spectral response and faceplate uniformity. The useful area of the plumbicon faceplate is a square about 1 cm on a side.

Previous work (1) has shown that an image consisting of 64,000 points, each point to be digitized to 64 gray levels, is sufficient for a wide spectrum of image processing and analysis. The 64,000-point raster is produced by a 256-line noninterlaced scan. The video signal in each line is sampled at 256 points and digitized to produce the required 256 x 256 $\simeq$ 64,000 picture element (pixel) raster. The sampled video of each pixel is quantized to 64 levels (6 bits) and stored in the 6 least significant bits of an 8-bit register. Two of these registers are provided to allow two successive pixels to be digitized and then assembled as one 16-bit word to be transmitted to the computer. With presently available direct memory access (DMA) devices, the data transmission rate is generally limited to 10^6 words/sec. Allowing for the usual retrace times, the standard TV scanning rate is about four times too fast. Thus, the camera was modified to scan the 256-line raster in $\frac{1}{15}$ sec, instead of the usual 525-line *interlaced* scan in $\frac{1}{30}$ sec. The actual data transfer rate across a line is 650,000 words/sec.

The 256 x 256 pixel raster is subdivided electronically into 16 x 16 pixel boxes and, in a given scan, any rectangular area made up of an integral number of those boxes can be selected and read into the computer. Thus, for a given application, images smaller than 256 x 256 points can be read in with a resulting saving of memory space. The location and dimensions of the rectangular area read into memory are, of course, under computer control. During horizontal and vertical retrace, and whenever the scanning beam is outside of the area to be digitized, the DMA transfer halts and the computer is freed for other processing. This allows effective multiprogramming within the standard RDOS environment.

The video signal is a linear representation of the brightness level at the face of the plumbicon, and, in the case of a transilluminated object (*e.g.*, a microscope slide), is proportional to the point-to-point transmission coefficient (T) of the object. For reasons of scaling and background subtraction, it is usually more convenient to digitize and process a signal that is proportional to the optical density D, where $D = -\log T$. For this reason SCANIT incorporates a logarithmic amplifier which can be inserted, under computer control, in the video chain just ahead of the A/D converter. Gain and offset controls on this amplifier adjust the resulting scales to give the same digitized values at the end points. The nominal range of the system is a factor of 100 in brightness, corresponding to a density range of 2. Typically, the light levels are adjusted such that in the transmission mode, transmissions of 1–100% are digitized linearly as transmission gray levels 0–63. In density mode, the corresponding optical densities 2–0 are digitized linearly as density gray levels 0–63.

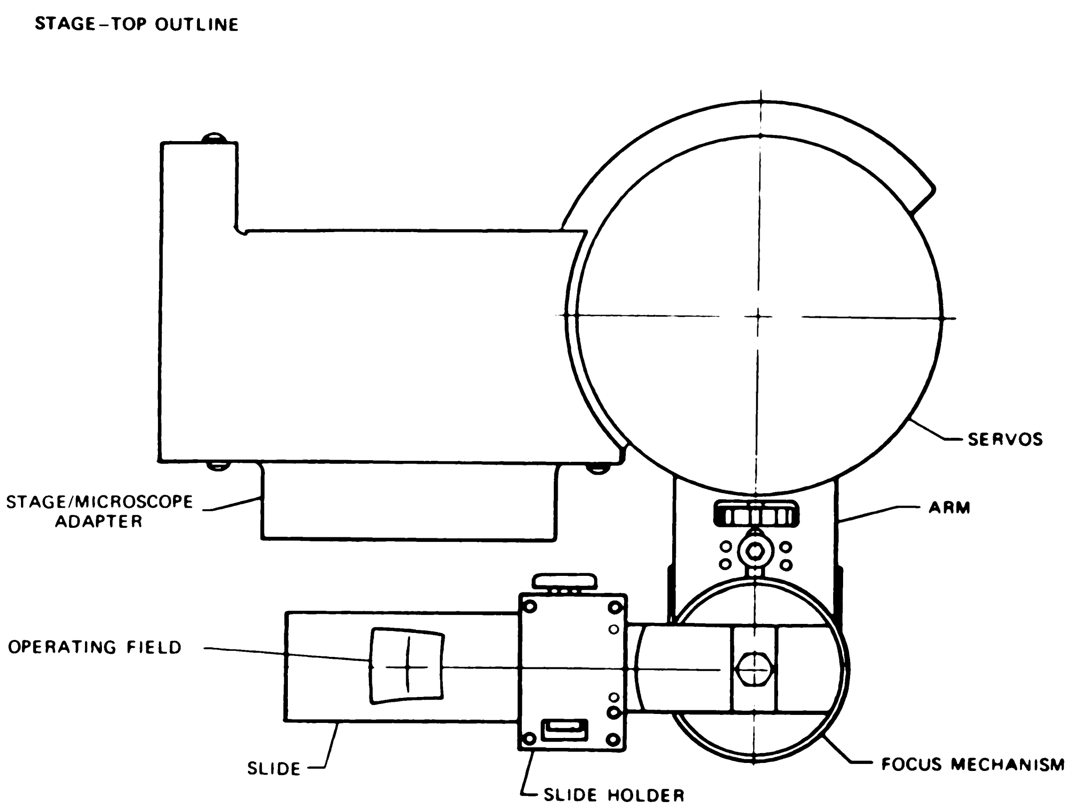

FIG. 3. Schematic top view of stage showing the double-pivoted structure and the quasi-Cartesian coordinate system.

Computer image processing techniques place a high premium on the equality of bin sizes in this analogue-to-digital conversion process. As a practical limit we have found that the analogue width of any level should deviate no more than $\pm 3\%$ (rms) from the average over all levels. In the parlance of A/D converter manufacturers, this property of having equal level widths is called "differential linearity." Typical A/D converters, however, are specified to have a differential linearity of "½ of a least significant bit." Thus, the 6-bit converter required for 64-level digitization would have a level width variation of $\pm 50\%$, about 16 times that desired. The solution to the problem is to use an A/D converter having more bits resolution than is needed for digitization. In SCANIT, therefore, a 10-bit A/D converter is used to obtain the 6-bit digitization. No time is wasted, however, since the converter is a successive approximation type and is "short-cycled," or stopped, after the 6 most significant bits are determined. The differential linearity is $\pm 50\%$ of a 10-bit level or $\pm 3\%$ of a 6-bit level, as desired.

With the $\times 100$ oil immersion objective generally used, the magnification from slide to plumbicon face is 200:1. Given that the 256 x 256 point raster covers a 1-cm square on the plumbicon face, the raster spacing works out to about 0.2 μm/pixel. This is comparable with the 0.25 μm Rayleigh resolution for good $\times 100$ oil immersion objectives. Lower power objectives can of course be used.

Although microscopic images are currently of the greatest interest, the TV camera can be removed from the SCANIT frame to capture virtually *any* image. Thus it is possible to digitize large films on a viewbox, *e.g.*, lung X-rays, or digitize large prints taken by scanning electron microscopes or even digitize "live" subjects such as human faces. This sort of versatility is not possible with CRT-driven and mechanical flying spot scanners.

Illumination system: Since the original design for SCANIT was reported (2), a second light source has been added, and there are now two sources for substage illumination. One is a standard quartz-halogen source for normal visual work, the other is a 150-watt high pressure Xenon arc source which has to be used when very narrow band illumination is required. A manually operated mirror is used to switch between sources. The Xenon arc source has a very uniform spectral response over the visible region. Large peaks present in the infrared region are eliminated by a "high pass" or infrared absorbing filter.

Also differing from the original design is the light filter system. The original was a linear filter system, having slots for four filters, mounted between the optical tower and the TV camera lens. A more versatile filter system was added below the substage condenser, as is shown in Figure 2. This is a rotary system consisting of a disc, holding up to eight 1-in diameter filters, driven directly by a stepper motor. The stepping rate is computer controlled to produce smooth starts and stops, and the system can make a smooth change from one filter to an adjacent one in about ⅕ sec. An important feature of this system is its placement between the light source and the slide. This ensures that light reaches the stained specimen only after being greatly attenuated by the filter, and this in turn minimizes fading. Exposed to the focused and unfiltered Xenon arc source, the illuminated area of a Wright-stained slide fades beyond recognition in less than 1 min. However, using a filter that reduces the light to a level just sufficient to produce saturation in the plumbicon, the same slide can be observed for hours without noticeable effects.

The effects of narrow band filtered light on cell images is shown for some typical white cells in Figure 4. Images at 600, 550, 490 and 420 nm are shown (10-nm band widths). The 600- and 490-nm images are the pair used to obtain color information from the Wright-stained cells, 600 nm being an orange color while 490 nm is perceived as blue. Three-color systems using tristimulus values are not needed, since the Wright stain is essentially a two-color dye system, eosin and methylene blue. The 600- and 490-nm bands are near the absorption peaks of the methylene blue and eosin dyes, respectively.

The 550-nm band (yellow-green) is between the two dye absorption peaks and records eosin and methylene blue absorption equally. Thus, no matter what the apparent color of the nucleus and cytoplasm, the nucleus is generally darker than the cytoplasm when seen through the 550-nm filter. This filter is therefore useful in separating nucleus from cytoplasm once the cell has been isolated from the background. The only cell type for which this fails is the basophil, in which the nucleus is usually masked by extremely heavily stained cytoplasmic granules.

Isolation of the cell from a background of plasma and red cells is accomplished in the 490- and 420-nm images. Histogram and boundary trace procedures (1), applied to the 490-nm image, produce a boundary which will include touching red cells in many cases. The red cells can be recognized and eliminated, however, because only *they* show up in the 420-nm (violet) image. This is because hemoglobin, present only in red cells, absorbs strongly at ≃420 nm, while the Wright-stain dyes are essentially transparent there.

Communication system: Communication between the human operator and the SCANIT system is provided by a TV monitor, a video terminal and a set of 32 push-buttons. The TV monitor is a modified 9-in CONRAC black-and-white studio monitor synchronized to the camera sweep. It can display images directly from the camera, before or after digitization, in log or linear mode, normal or black-white inverted (negative image). These are called *copy* displays and do not require data transfer to or from the computer. The monitor can also be used to display images from the computer's memory. These may be modified scanned images, with marked boundaries as in Figure 5, or they may be text or graphical displays generated within the computer. These displays from memory are refreshed 15 times/sec by DMA transfers, however, and therefore consume considerable processor time. For this reason, text and graphical displays are usually transmitted to the video terminal screen, where refresh is via the terminal's internal storage and does not involve extensive data transfer from the computer.

The 32 computer-readable buttons are used by the operator to control branch points in programs, and required alphanumeric strings are entered via the video terminal keyboard.

One should also mention the high speed line printer as a communication device both for text and for images. As can be seen in Figure 6, it provides hard copy output of cell images using specially adapted overprinting techniques to obtain the required contrast. On these images, 10 gray levels are distinguishable.

PRELIMINARY EVALUATIONS

The SCANIT system as a whole has been operating for seven months and most of the "bugs" which plague any new hardware design have been dealt with. Most initial problems were in the digital design and involved the interfacing of SCANIT to the NOVA 840 in a real time, multiprogrammed system environment. No really serious difficulties arose, however, and SCANIT now operates stably under the fully mapped RDOS operating system with minimal interference with other users.

Stage cell-finder performance: In assessing the performance of the stage it is useful to relate its specifications to a concrete example of cell finding, namely, finding leukocytes in peripheral blood smears. In the automatic cell-finding mode, the stage is designed to slew at about 120,000 steps (30,000 μm)/sec. The search aperture for the photodiode sensor is equivalent to a circular disc 8 μm in diameter in the object plane. With this aperture and slow speed, and allowing for the controlled acceleration starts and stops, the system is designed to start up, find, stop and center a leukocyte in about 1/2 sec on a normal peripheral blood smear. Allowing 1/2 sec for processing, the SCANIT system could handle 1 cell/sec as a fully automatic differential leukocyte counter. Although this is comparable to the usual manual counting rate, the SCANIT system is not expected to compete with commercially produced differential counter systems in terms of speed. For these specialized systems, wide area low-resolution search techniques, multiple video channels, and hardwired processing could permit nearly a factor of 10 increase in speed. SCANIT, by comparison, avoids specialization and remains flexible by using only software processing, and it is estimated that the speed of the present cell finding system matches or exceeds this processing capability in any of the contemplated research applications.

At present the cell-finding system has been operated successfully only at one-half of its design speed. At that speed it does find better than 98% of the leukocytes present, and false stops, with no dark object in the field, occur less than 5% of the time. Cell centering is sufficient to bring essentially all cells completely within a 36-μm square box (180 x 180 pixels). The finder also stops at large stain crystals, large platelet clumps and at other large dark objects as expected. At full design speed, too many cells are skipped and cell centering is inadequate. This appears to be a noise problem related to

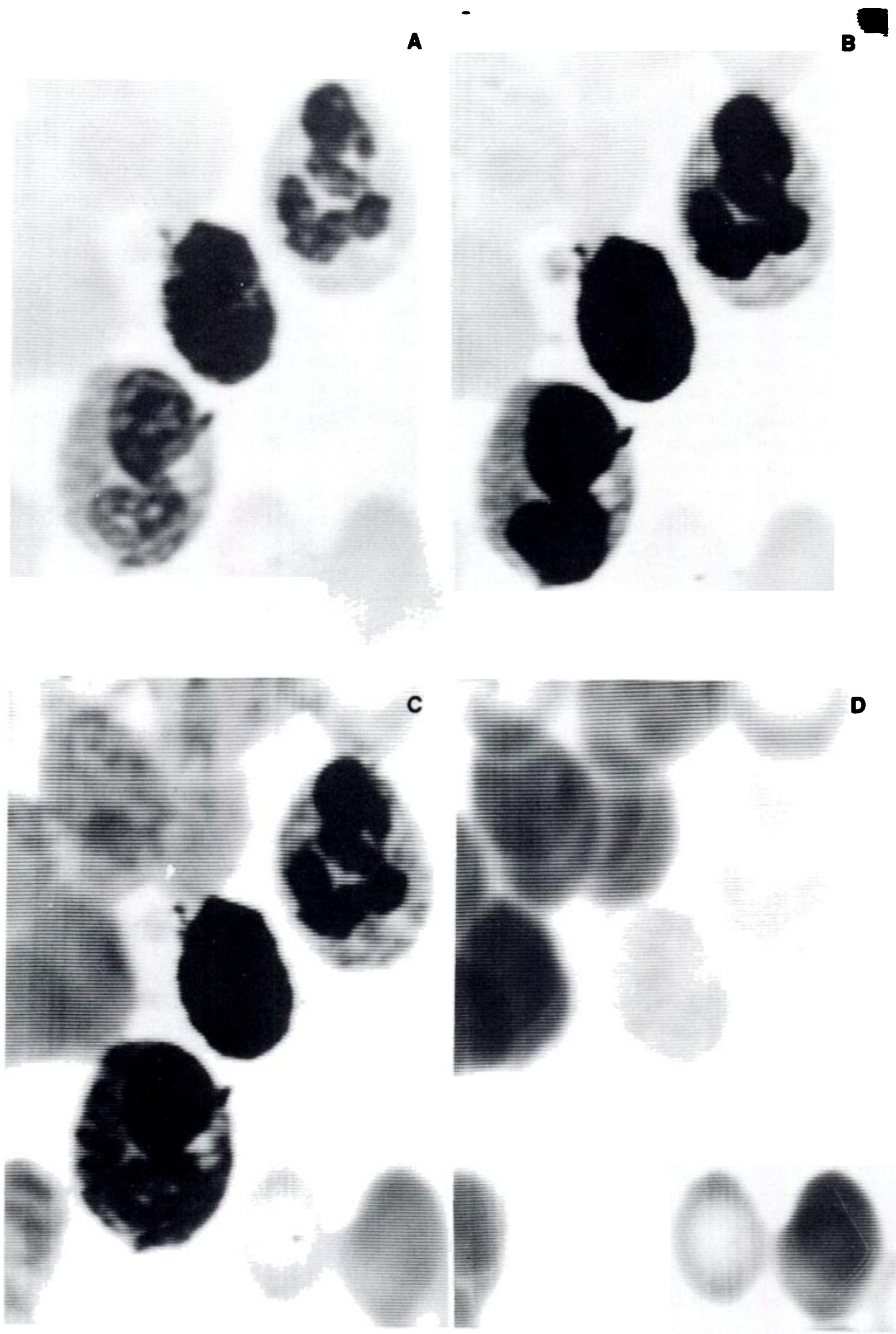

FIG. 4. Three leukocytes against a background of erythrocytes, as seen by SCANIT, showing the effects of narrow band (10 nm) illumination at several wavelengths. (A) 600 nm (orange); (B) 560 nm (yellow-green); (C) 490 nm (blue); (D) 420 nm (violet). Greatest nuclear/cytoplasmic contrast is shown in the yellow-green image (B). The delicate pink color of the neutrophil cytoplasm (top-most leukocyte) and of the erythrocytes is rendered light in the orange image (A) and moderately dark in the blue image (C). In the violet image (D) the leukocytes are very faint (even the relatively dark basophil in the center), and the erythrocytes are rendered dark because of hemoglobin absorption at 420 nm. All photographs are taken from the SCANIT monitor screen. Images were linearly digitized (transmission gray levels) and reconverted for display (digital copy mode).

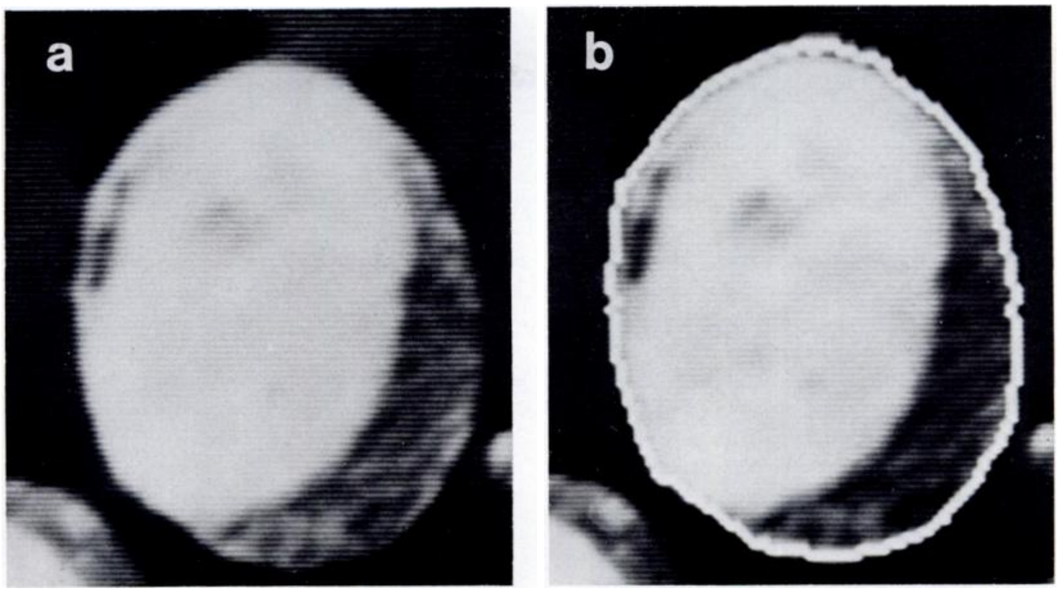

FIG. 5. A promyelocyte image processed to find the cell boundary. a, raw image; b, image with cell boundary marked in white. Both photographs are taken from the SCANIT monitor screen. Illumination was through the 560-nm (yellow-green) filter. The raw image was read from disc, displayed, processed and marked, then displayed again. A negative image was used to emphasize the light boundary.

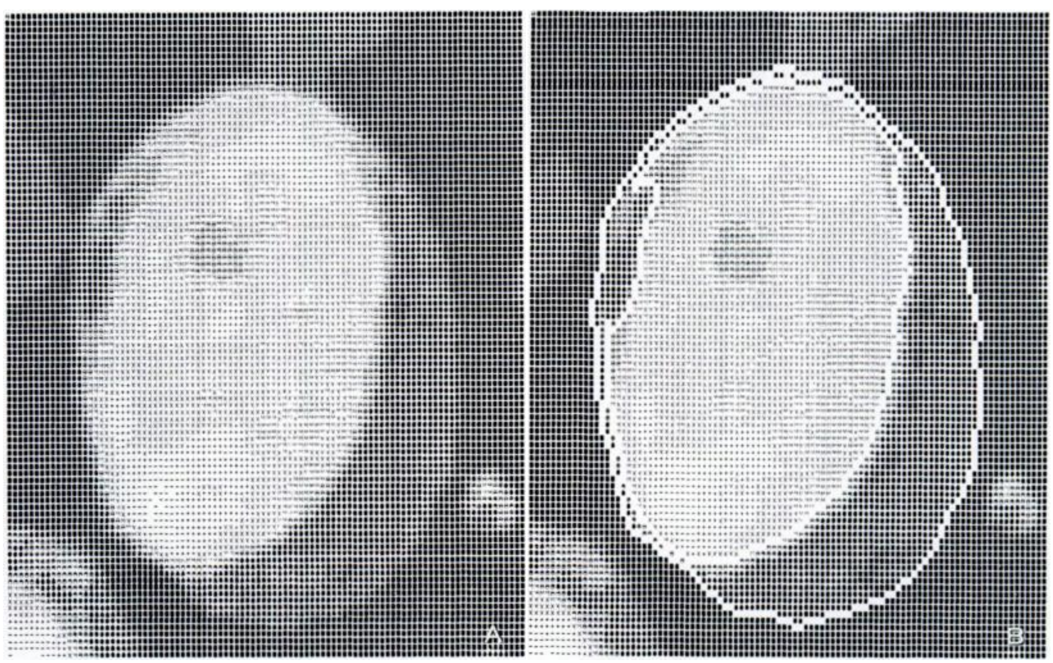

FIG. 6. Line printer images of the cell of Figure 5. A, raw image; B, image with marked cell and nuclear boundaries.

the servo system band width. Modifications and further testing are planned.

Scanner/digitizer performance: In evaluating the performance of the scanning system the most important questions, to which no *a priori* answers were available, were what spatial resolution, field uniformity and video signal-to-noise ratio could be achieved.

Spatial resolution will only be tested fully after a custom-made, etched chromium microscope resolution target is completed. Rough measurements made with stage micrometer scales having 1-μm wide lines indicate an overall modulation transfer function (MTF) of perhaps 30% at 500 line pairs/mm (500 lp/mm) referred to the slide plane. Practically speaking, spatial quantization limits the measurement of MTF to the smallest line spacing that guarantees one pixel being entirely within a given line. Thus, the 0.2-μm raster spacing prohibits MTF measurement at more than 1250 lp/mm. Testing scanned white cell images, it has been observed that the 0.5-μm granules which are often present in the cell cytoplasm are being resolved satisfactorily.

Field nonuniformity has essentially two components. First there is a low spatial frequency component, or *shading*, due to nonuniform illumination of the slide and to the plumbicon scan geometry. Then there are higher frequency effects due to local imperfections. Because of the highly collimated nature of the light emanating from the trinocular tube, every lens bubble, coating flaw, granularity and dust speck will show up as a high-frequency nonuniformity. These effects can be lumped together with plumbicon faceplate blemishes and be simply called *speckling*.

Shading is shown graphically in Figure 7. Except at the very edges of the raster, where an image mask interferes, the worst variation from the mean is 1.25 density levels (about 10%). In

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	-0.19	-0.17	-0.17	-0.19	-0.20	-0.29	-0.63	-0.98	-0.93	-1.15	-1.23	-1.36	-1.34	-1.64
2	-0.26	-0.17	-0.14	-0.16	-0.15	-0.18	-0.27	-0.26	-0.47	-0.66	-0.84	-1.17	-1.18	-1.25
3	-0.19	-0.17	0.06	0.27	0.03	-0.13	-0.16	-0.13	-0.19	-0.19	-0.39	-1.05	-1.16	-1.20
4	-0.18	-0.04	0.58	0.64	0.54	0.40	0.19	-0.07	-0.18	-0.18	-0.21	-0.60	-1.09	-1.47
5	-0.17	0.14	0.79	0.82	0.75	0.67	0.08	-0.14	-0.14	-0.17	-0.19	-0.35	-0.86	-1.19
6	-0.14	0.34	0.82	0.80	0.72	0.71	0.37	0.16	-0.12	-0.16	-0.15	-0.18	-0.34	-0.46
7	-0.12	0.69	0.82	0.82	0.80	0.79	0.81	0.48	0.01	-0.15	-0.16	-0.18	-0.19	-0.19
8	-0.14	0.52	0.86	0.89	0.84	0.83	0.77	0.75	0.22	-0.16	-0.17	-0.18	-0.19	-0.27
9	-0.07	-0.05	0.58	0.84	0.84	0.83	0.78	0.59	0.19	-0.12	-0.16	-0.19	-0.18	-0.18
10	0.18	0.11	0.08	0.75	0.83	0.80	0.63	0.47	0.35	-0.16	-0.16	-0.16	-0.17	-0.16
11	0.28	0.68	0.67	0.76	0.81	0.79	0.57	0.57	0.39	-0.16	-0.17	-0.18	-0.20	-0.19
12	0.46	0.46	0.50	0.61	0.76	0.72	0.64	0.14	-0.12	-0.16	-0.18	-0.19	-0.21	-0.21
13	0.06	0.10	0.01	0.08	0.25	0.11	0.09	0.02	-0.14	-0.16	-0.18	-0.19	-0.16	-0.10
14	-0.18	-0.36	-0.56	-0.16	-0.13	-0.11	0.01	0.18	0.24	-0.15	-0.17	-0.17	-0.07	0.20

FIG. 7. Graphic display of shading information for SCANIT. A blank field (clear slide) was mounted and defocused to minimize the effects of dust, etc. Average *density* gray levels were then calculated for each of the 16 x 16 pixel boxes within the 256 x 256 pixel raster. Number shown for each box is the difference between the average density gray level of that box and the "grand mean" for the entire image. The outermost boxes were not scanned, since an image mask on the plumbicon face is visible in these boxes. This mask, of course, is not relevant to shading calculations. The inscribed box indicates the centered 96 x 96 point area often used for cell images.

the centered 96 x 96 point raster that we often use, it is about 0.5 density levels at worst. Speckling is also less than one level if the optics are clean. By recording blank-field reference images every few hours, it has been possible to correct for all of the image nonuniformities by simple image subtraction using density gray levels. Although not usually necessary, this correction is good to better than one gray level in all cases, even including those where moderate speckling causes 4–5 level blemishes in the uncorrected image.

To digitize to 64 levels with an accuracy of ±1 level (rms) in the optical density mode requires a peak-signal/rms-noise figure in the raw video substantially greater than 64:1, because the log amplifier used has a high incremental gain at the low light or high density end of the scale (see Fig. 8). In fact, to obtain an rms deviation of one level in this region requires a peak-signal/rms-noise ratio (in raw video) of about 1300:1. Partly because the slow scanning speed and faceplate conductance limit the signal current, SCANIT obtains a figure closer to 250:1, and the desired ratio can only be obtained by scan averaging. Averaging eight scans gives an rms noise of 1.0 levels at density level 10, and the light level is typically adjusted such that little or no noise-sensitive picture information is below level 10.

High speed assembly language subroutines have been written to do scan averaging. With these routines SCANIT typically has been scan averaging eight images of size 96 x 96 pixels for each of six filters, and recording the six averaged images on disc in about 30 sec.

Automatic focus evaluation: In the original design of SCANIT (2), a system of hardware

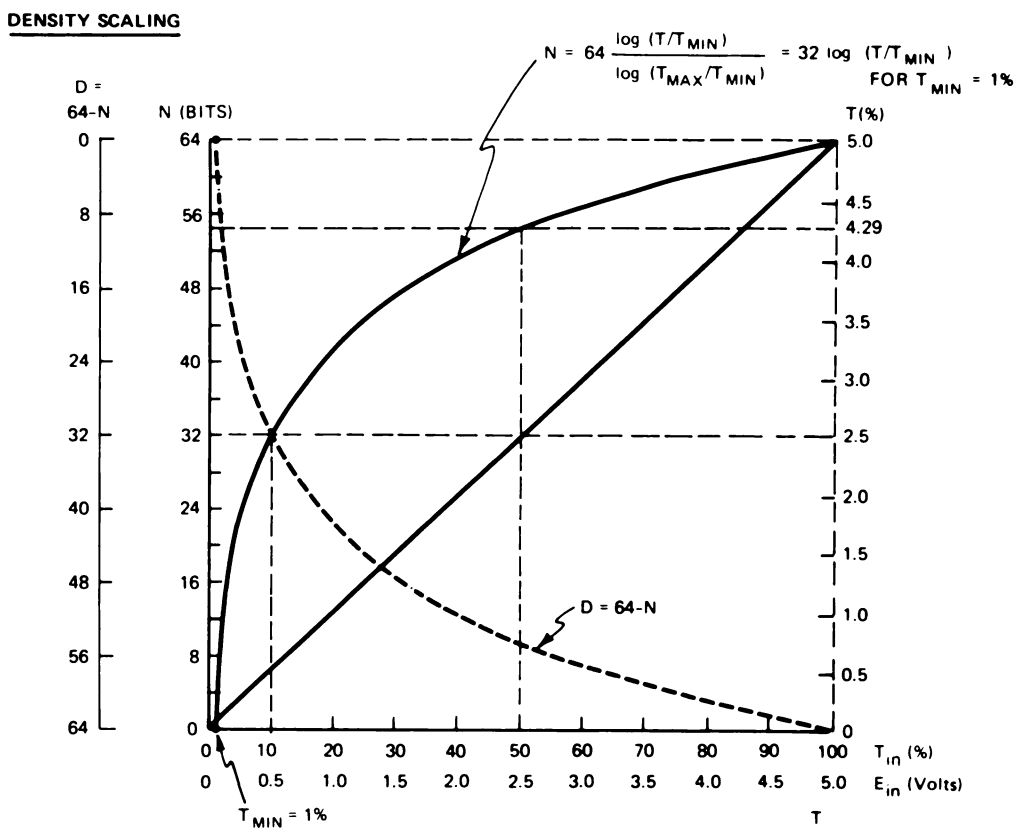

FIG. 8. Log amplifier response as a function of input voltage. Scales show the relationship between density and transmission gray levels as well as voltage relations. The dotted curve is optical density (D) versus transmission (T). The solid curve is "inverted density" (N), which is the logarithmic response corresponding to transmission, in that high N values correspond to high T. The steep slope near the low end of the scales corresponds to high incremental gain in the log amplifier and produces noise amplification. This results in a lower than desirable signal/noise ratio at the low end.

automatic focus was proposed. At present, final implementation of this system has not been completed, but a software simulation of the proposed method has been successfully developed. A focus function $f(Z)$ is calculated which is a measure of the average change in gray level between pairs of points separated by n pixels. $f(Z)$ is a maximum when the image is in focus, and is given by:

$$f(Z) = \sum_{j} \sum_{i} \{G_i(Z) - G_{i+n}(Z)\}^2$$

Where the index (i) ranges over all image points, in order, along a scan line (j); n is a small integer; Z is the Z-axis, or focus position; and G_i is the transmission gray level for point i. A value of n equal to 2 gives a good signal-to-noise ratio, and a plot of $f(Z)$ versus Z for a typical cell image is shown in Figure 9. An algorithm has been developed that finds the peak of this curve in an average of six scans, taking about 3 sec total time. This same algorithm using the proposed hardware-generated focus signal could find the peak in the same time taken for the six scans, namely $6/15$ sec. This algorithm focuses repeatably to less than one 0.25-μm step, which has been found to be about as good as a human operator.

DISCUSSION

The initial results with SCANIT indicate that the plumbicon scanner system, with computer corrections for nonuniformities, and scan averaging, can produce images of sufficiently high quality to permit the study of a variety of image processing and automation techniques. It also has the high speed which will, for example, permit examination of many thousands of cell images, each at several wavelengths, during a year's investigation. For medical research applications, where flexible multipurpose machines are to be preferred over special instruments intended for one particular image-processing task, SCANIT appears superior to flying spot

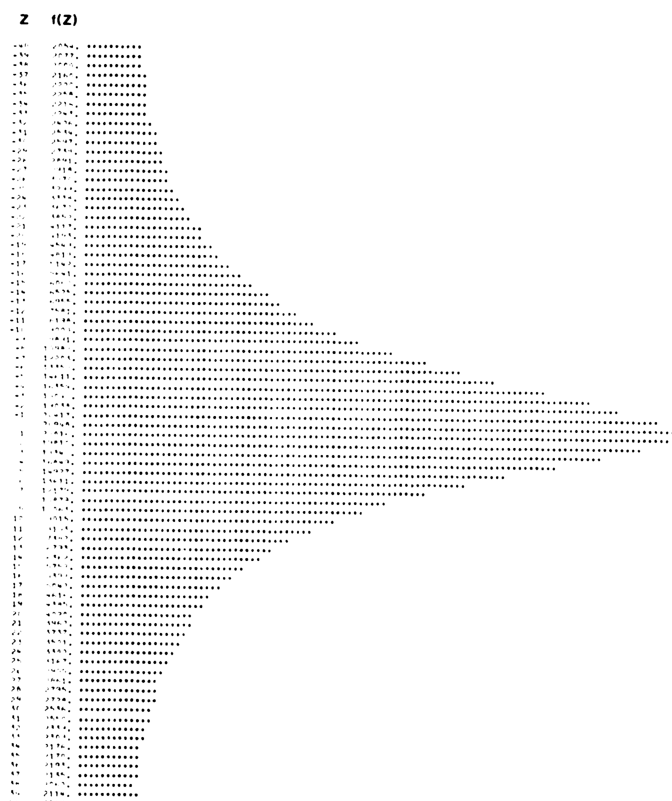

FIG. 9. Typical focus signal $f(Z)$ as a function of Z. Z is measured in 0.25-μm steps, with best focus at $Z = 0$. Image used was of a metamyelocyte, but the shape of the $f(Z)$ is essentially invariant to a change of cell type.

and mechanical scanners. The authors feel that its balance of speed, cost, versatility and image quality make it the scanner of choice for these applications, with the limitation of scanning a maximum of 64,000 points at one scan.

The stage, cell finder and focusing systems have proved adequate for the research purposes presently contemplated, but further developments are planned to increase their speed for future projects.

ACKNOWLEDGMENTS

The authors would like to express their appreciation to James Falzano and Nunzio Sansevero at the C. S. Draper Laboratory and to Kenneth Yasi at Tufts-New England Medical Center for their help in "debugging" and modifying the SCANIT system, and to Ben Weintraub at Tufts for his help in writing the boundary finding and display programs.

LITERATURE CITED

1. Brenner JF, Gelsema ES, Necheles TF, Neurath PW, Selles WD, Vastola E: Automated classification of normal and abnormal leukocytes. J Histochem Cytochem 22:697, 1974
2. Dew B, King T, Mighdoll D: An automatic microscope system for differential leukocyte counting. J Histochem Cytochem 22:685, 1974